

Suniksha Gupta Seattle, WA | sunikshagupta695@gmail.com | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Aspiring **Data Scientist** with experience in predictive modelling, **EDA**, and **machine learning** using **Python** and scikit-learn. Strengths include analytical thinking, statistical skills, and a solid research background (**Ph.D. in Physics**). Driven to apply data-driven methods to real-world problems, bringing strong modelling ability, rigor, and fast learning to data science roles.

SKILLS

Programming Languages: Python, SQL, MATLAB, C (basic), HTML (basic)

Tools & Libraries: NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, Jupyter Notebook, Google, Git/GitHub, MS Office

Machine Learning: Linear Regression, Random Forest, Gradient Boosting, K-Nearest Neighbors, Decision Trees, SelectKBest, PCA, Feature Engineering, Cross-Validation, Model Evaluation

Data Science Techniques: Exploratory Data Analysis (EDA), Data Preprocessing, Data Cleaning, Train/Test Split, Hyperparameter Tuning, Visualization

Deep-Learning Frameworks: TensorFlow (basic), Keras (basic)

WORK EXPERIENCE

Assistant Professor (Research Cell) | Poornima University, Jaipur

Aug 2023 – Jan 2025

- Delivered 20+ hands-on technical sessions on PVD and thin-film fabrication, focusing on experimental planning, data collection, and statistical analysis.
- Taught and supervised 100+ undergraduate students, guiding them in lab experiments, measurement techniques, and interpreting scientific datasets.
- Led research on thermoelectric thin films, producing 2 peer-reviewed publications and presenting findings at 2 national conferences.

Research Fellow | Department of Physics, MNIT Jaipur

Jan 2018 – Dec 2022

- Conducted research on nanomaterials and thermoelectric thin films, generating and analysing experimental datasets to study performance trends.
- Optimized thin-film synthesis protocols (PVD) and applied statistical methods to evaluate material properties and improve efficiency.
- Published 7+ peer-reviewed papers and presented research at national conferences and workshops.

Research Intern | CSIR–National Physical Laboratory, New Delhi

Jan 2016 – Aug 2016

- Experimental studies on thin-film materials, analysing electrical and structural datasets.
 - Assisted in building theoretical prediction models for Resistive RAM behaviour using parameter-based data trends.
 - Performed 100+ measurements (I–V, sputtering parameters), generating structured datasets for analysis.
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Selected Data Science Projects:

- **Capstone 1: Predicting Thermoelectric Efficiency using Machine Learning** — Developed regression models (Random Forest, SVR) to predict ZT (Figure of merit) of silver chalcogenide-based thermoelectric materials. Cleaned and pre-processed public datasets (~2000 entries) using pandas and scikit learn. Applied **feature scaling, cross validation and hyperparameter tuning** to improve model accuracy by 12%.
 - **Capstone 2: Predicting House Prices** — Building and evaluating **supervised machine learning models** to estimate real estate values based on features such as location, square footage, and amenities. Progressing through advanced modules on **predictive analytics, data visualization, and storytelling** to develop deployable ML solutions.
 - **Capstone 3: Predicting Ticket Prices for Big Mountain Resort** — Developed ML models using scikit-learn to predict ski resort ticket prices based on features like vertical drop and night skiing capacity. Performed EDA and feature selection (SelectKBest) to identify key predictors and reduce multicollinearity. Compared **Linear Regression and Random Forest**, achieving improved accuracy with the Random Forest model.
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CERTIFICATIONS

- IBM Data Science Professional Certificate (Coursera)
 - Machine Learning — Andrew Ng (Coursera)
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EDUCATION

Springboard USA

Data Science

Online course

Dec 2025

Malaviya National Institute of Technology, Jaipur

Rajasthan, India

Ph.D. in Physics (Material Science)

Aug 2023

Punjabi University Patiala

Punjab, India

Master of Science in Physics

Aug 2016